

Mobile App (Continuous Runner) Design Document

Bryce Dixon (3397649) Dec 2025

Game Title: Christmas Miracle

Backstory:

Our main character, Buddy the Bunny, works at the North Pole in helping to prepare gifts for Santa. While working in the workshop, Buddy caught an evil monster (Ungiftus) pouring magic dust on the huge pile of gifts that Buddy had completed that year. This dust teleported the gifts all around the world. Now it is up to Buddy to travel around the world and collect them as efficiently as possible, while avoiding Ungiftus' army.

Characters:

- Buddy → A bunny that runs.
- Ungiftus's army → These are individuals walking around the level that Buddy must avoid and/or shoot (with snowballs).
 - Dragon → Smaller enemy which wanders and can be jumped on, but not shot.
 - Skeleton → These dance up and down, and can be shot with snowballs.
 - Floating rock → This hovers back and forth, can be shot with snowballs.

Gameplay:

- Gameplay Mechanics / GameFlow:
 - This is a continually scrolling runner game.
 - The user controls Buddy to either "Jump" or "Shoot".
 - The maps are predetermined, and the map scrolls from right to left automatically.
 - So the player appears to be continually running to the right.
 - The user can collect:
 - Their gifts
 - Throwing upgrades (faster firing rate for snowballs).
 - Jumping upgrades (jump higher and further for 4 seconds).
- Rules for the Game:
 - The player's objective is to get through the levels and collect as many gifts as possible.
 - Currently there are 3 levels, which link to one another.
 - Once you finish the 3rd level, you are looped back to the first level and continue playing.
 - This is done until the player loses.
 - If they collide with a monster or fall off of the platforms, they lose.
- Victory Conditions:
 - Collect as many gifts as possible without dying. Once game over occurs, the score is the total amount of gifts gathered. If the total amount of gifts is more than the high score, it is saved as the high score. If it is equivalent to the high score, but was done in less time, it will be the new high score.

- Time is simply gathered as a tie breaker so that the primary motivation is gift collection, but time plays a role for the user to try and optimize their paths for maximal gift collection in the time.

Game World:

- **BUTTONS:**
 - RESTART → this restarts the entire game play at level 1.
 - PAUSE → this pauses the screen.
 - JUMP and SHOOT → Make the player jump or throw snowballs.
- **PLAYER (Buddy):**
 - Buddy runs using a spritesheet.
 - When buddy is "shooting" at the enemies, it appears as if they are throwing a snowball.
- **ENEMIES:**
 - Dragon (small), Skeleton (medium size), floating rock (largest)
 - When any of these enemies are killed, their images change and they dramatically fall down the screen and out of view or fly off to the right.
 - The dragon is jumped on.
 - The skeleton and rock are shot with snowballs.
 - Spikey ice walls are also in the level and must be shot.
- **BACKGROUND/ TILES:**
 - Each background is quite simple overall with a single color, as a gradient background was too distracting.
 - Every second level has wooden tiles and a green background, while odd level have snow platforms and blue background.
 - Moving tiles are always snow platforms.
- **LEVELS:**
 - There are 3 levels linked together.
 - When you finish level 3, it loops back and starts at level 1. This way the player can play continuously and strategically looping through the levels, gathering upgrades and gifts to get the high score.

BONUS MATERIALS:

- The motivators for the game will simply be gift collection and time.
- So the user will want to try to get through the level effectively, choosing the path that gets them the most gifts.
 - Some levels have alternative routes, so the user can choose the route with more gifts, although if you make a mistake trying to get more gifts, it is much harder to survive the level.

FUTURE FEATURE IDEAS:

- Make more levels or increase level speeds as it loops.
- Add more enemies, which populate the levels so that each time the user comes back to a level, it may feel new.

- Add other upgrades:
 - Different weapons.
 - Shield (with a timer)
 - Enemies that track the user and not just pace or move on the spot.
 - Enemies that are harder to kill (take multiple snowballs).

Thank you for reading!

Mobile App Test Plan

Bryce Dixon (3397649) Dec 2025

Test Item	Input	Expected Output	Actual Output	Results/Remarks
Display (Testing item animations/ behaviours)	Present	Present should float and appear to shrink and grow on the spot.	Presents floats, while shrinking and growing.	COMPLETE.
	Buddy	Buddy should appear to run to the right, stay in the left-middle of the screen.	Buddy runs to the right and stays in the middle of screen-left.	COMPLETE.
	Floating Rock	It should travel left and right to a max of 4 spaces, appearing to hover up and down.	Hovers up and down and travels back and forth on its platform to a max size of 4 spaces.	COMPLETE.
	Skeleton	It should travel up and down 2 spaces while dancing.	Moves up and down, and dances.	COMPLETE.
	Dragon	Should travel to the right and left, to a max of 5 spaces, appearing to walk.	Walks left and right on its platform to a max of 5 spaces.	COMPLETE.
	Horizontal tile	Tile should move left and right 2 spaces.	Tile moves left and right 2 spaces.	COMPLETE.
	Vertical tile	Tile should move up and down 2 spaces.	Tile moves up and down 2 spaces.	COMPLETE.
	1. Snow and wood Tiles, 2. Jump upgrade 3. Teleport door 4. Snowball upgrade 5. Spikey ice walls	These should all appear as is without any animations.	All items appear correctly on screen as per their images.	COMPLETE.
Screen buttons	Press Restart Button	All scores, enemies, collectables and time	Everything resets to the beginning and	COMPLETE. Works when pressed on

		should reset, player should reset at the beginning of level 1.	gameplay fully restarts from level 1.	any level. Always returns to level 1.
	Press Pause Button	1. When press pause during gameplay, the pause screen should appear and the game should pause. 2. When press pause on pause screen, gameplay should continue.	Pause screen appears when press pause during gameplay, pressing pause again continues gameplay with all variables and playing conditions unchanged.	COMPLETE. Pause button works as expected.
	Press Jump Button	1. When pressed, player should appear to jump up and come back down on its own. 2. The player should not be able to jump again while in the air.	The player jumps the same amount each time. When the player is in the air, the jump button does nothing.	COMPLETE. Jumping functionality is working as desired.
	Press Shoot Button	When pressed, a snowball should appear from the player and travel across the screen until it collides with an enemy or tile.	The snowball appears from the player, and travels across the screen as expected. Works when player is jumping in the air as well.	COMPLETE. Shooting works to the level needed for gameplay.
	Touch throughout the rest of the screen	1. During gameplay, the user should be able to press anywhere else on the screen with no affects. 2. Pressing on screen during paused gameplay, only pause and restart buttons should work.	1. Pressing anywhere else on the screen has no affect on the game. 2. In the pause screen, only the pause and restart buttons work.	COMPLETE. Works as expected.
Collectibles/ Upgrades/ Floor Tiles	Run into Present	The present should disappear, the HUD should increment the present counter by 1.	Present disappears and present counter increments by 1 in the HUD.	COMPLETE. Present collection works.

	Run into Snowball Upgrade	The icon should disappear, the HUD should increment the snowball rate by 1, and the player's rate of fire should increase when "shoot" is pressed.	Icon disappears, HUD increments appropriately, player's firing rate increases when shoot button is pressed.	COMPLETE.
	Run into Jump Upgrade	1. The icon should disappear, 2. a timer should appear in the HUD in larger yellow font, 3. the player should be able to jump twice as far. 4. Once the timer is up, the player's jumping should return to normal.	1. The icon disappears. 2. A timer appears in yellow in the HUD. 3. The player's jumps are twice as far. 4. After the timer, the jumping returns to normal.	COMPLETE. The timer is noticeable for the user.
	Run into Level Teleport Door	1. When the player hits the door, the game should stop, load the next level, then the next level should be playable in full. 2. The HUD and player state should be unchanged from the previous level.	1. Once hitting the door, the game stops, next level loads and pause screen appears. 2. Can continue play in the next level. 3. The HUD and player state carries between levels to maintain player state.	COMPLETE. Player state is consistent between levels.
	Snow tiles	Player and enemies should be able to walk on/ detect these.	Player and enemies are able to walk on these tiles.	COMPLETE.
	Wooden Tiles	Player and enemies should be able to walk on/ detect these.	Player and enemies are able to walk on these tiles.	COMPLETE.
Player/ Enemy Collision Detection	Buddy - dragon collision	1. Running collision: if buddy runs into the dragon from the left, it is game over. 2. If buddy is	1. When buddy is running and hits the dragon, game over occurs. 2. When buddy is jumping, if they collide	COMPLETE. Periodically there was a lag and buddy could walk through the dragon (only if there are

		jumping and hits the dragon, the dragon flies off of the screen.	with the dragon, the dragon flies off the screen.	multiple collisions at once). I modified buddy's hit boxes (lower on right and narrower bottom), and made the dragon slightly wider, which seems to have resolved this.
	Buddy - skeleton AND Buddy - floating rock collision	When buddy collides with a skeleton or floating rock, game over should occur.	Game over occurs when buddy collides with skeleton or floating rock enemies.	COMPLETE.
	Buddy - Spikey wall collision	When buddy hits a spikey wall, game over should occur.	Game over occurs when buddy hits the ice wall.	COMPLETE.
	Skeleton-dragon-floating rock collision	These characters should be able to move past one another without issue.	When the enemies collide with one another, no issues arise, and they continue unaffected.	COMPLETE.
Snowball Collision Detection	Snowball collision with collectables/ upgrades.	The snowball should travel through all collectables (jump upgrade, snowball upgrade, and presents)	Snowballs travel through each of these items. This is controlled in the VIEW class.	COMPLETE.
	Snowball - Tile collision	If a snowball collides with a tile, it should stop and disappear.	When a snowball hits the tile, it disappears.	COMPLETE. Works as expected.
	Snowball - dragon collision	When a snowball hits a dragon, it should disappear, but do nothing to the dragon.	The dragon is hard to hit with the snowball as it is short, but if it does, the snowball disappears and dragon has no changes.	COMPLETE.
	Snowball-skeleton collision	If a snowball hits the skeleton, its image should change to show it lying down, then it should float	The skeleton falls over and floats downwards off the screen when hit with a snowball.	COMPLETE.

		downwards off the screen.		
	Snowball-floating rock collision	The floating rock should swing its fists in the air and float down off the screen when it is hit with a snowball.	The floating rock swings its fists and floats off the screen when hit with a snowball.	COMPLETE.
	Snowball-spike wall collision	When a snowball collides with the ice wall, the ice wall should disappear.	The ice wall disappears when a snowball hits it.	COMPLETE.
Level Linking/loading	Level loading	When a level loads, it should load the level entirely including all collectables, enemies and tiles.	When levels load, all items are reset and loaded entirely.	COMPLETE.
	Levels should link in a continuous loop	Level 1 should link to level 2, level 2 to level 3 and level 3 to level 1.	Levels link as designed (1 to 2, 2 to 3, 3 to 1) to allow for continuous gameplay.	COMPLETE.
	Transition between levels should maintain player state.	When transitioning levels, player state should be maintained from the previous level.	Player state, including the HUD, player actions (shooting upgrade), time and score are maintained between levels.	COMPLETE.
Game Over	Game over can be triggered by:	1. Collision with dragon, floating rock, skeleton, or ice wall. 2. Falling off the platforms.	1. Game over is triggered when collides with enemy or ice wall (tested above). 2. Game over is triggered when falls below the map.	COMPLETE.
	When game over is triggered.	1. The game over screen should appear, 2. game play should stop. 3. The player's score and time should display.	1. Game over screen appears. 2. Game play stops when this occurs. 3. The player's score and time appear. 4. The high score is displayed.	COMPLETE.

		4. The high score should display. 5. If the player gets a new high score, the system should display this to the user. 6. Hitting restart should reset everything and begin the user at level 1.	5. A new highscore is indicated in yellow writing. 6. Hitting restart resets the user to begin anew from level 1.	
Different Screen Sizes	Medium Phone	Display should look the same on all devices.	Formats well and levels are playable on this size. HUD displays correctly and feels appropriate for the screen size.	COMPLETE.
	Foldable phone		Very similar to the medium phone, playability unchanged. HUD displays correctly and feels appropriate for the screen size.	COMPLETE.
	Tablet		The buttons appear slightly larger, and the player has to jump to shoot higher ice walls, but this does not impede gameplay. HUD displays correctly and feels appropriate for the screen size.	COMPLETE.
Sounds	Shoot	The shooting sound should play when shooting.	Unique shooting sound plays.	COMPLETE
	Jump	A jumping sound should play when jumping.	Unique jumping sound plays.	COMPLETE
	Snowball upgrade	A unique sound should play when upgrade collected.	Unique snow upgrade sound plays.	COMPLETE
	Jump	A unique sound	Unique jump upgrade	COMPLETE

	upgrade	should play when upgrade collected.	sound plays.	
	Gift collection	A unique sound should play when upgrade collected.	Unique gift collection sound plays.	COMPLETE
	Buddy runs into dragon, skeleton, floating rock, or ice wall.	A sound should play when collision occurs.	Unique end of game sound plays.	COMPLETE
	Teleport	A sound should play when collide with door.	The teleport sound plays when get to the teleporter and also when user gets to the new level.	COMPLETE
	Enemy (wall, skeleton, rock) is hit with snowball	A sound should play when enemy hit with snowball.	An explosion sound plays when a snowball hits any of these three objects.	COMPLETE
	Jump on dragon	A sound should play when buddy jumps on a dragon.	A unique sound plays when jump on the dragon.	COMPLETE
	Snowball hitting tile	A sound should play when the snowball hits a tile.	The unique snow sound plays when snowball hits a tile.	COMPLETE